Convex Optimization 作业 3

October 2025

1 Solve following LP by Two-Phase Simplex Tableau

$$\text{max. } 3x_1 - x_2$$
$$\text{s.t. } \begin{cases} 2x_1 + x_2 \geq 2 \\ x_1 + 3x_2 \leq 2 \\ x_2 \leq 4 \\ x_1, x_2 \geq 0 \end{cases}$$

1.1 Phase-I

$$\text{max. } -a_1$$
$$\text{s.t. } \begin{cases} 2x_1 + x_2 - s_1 + a_1 = 2 \\ x_1 + 3x_2 + s_2 = 2 \\ x_2 + s_3 = 4 \\ x_1, x_2, s_1, s_2, s_3, a_1 \geq 0 \end{cases}$$

	z	a_1	x_1	x_2	s_1	s_2	s_3	
	1	1	0	0	0	0	0	0
s_1	0	1	2	1	-1	0	0	2
s_2	0	0	1	3	0	1	0	2
s_3	0	0	0	1	0	0	1	4

R0-R1 使得基变量系数为 0

	z	a_1	x_1	x_2	s_1	s_2	s_3	
	1	0	-2	-1	1	0	0	-2
s_1	0	1	2	1	-1	0	0	2
s_2	0	0	1	3	0	1	0	2
s_3	0	0	0	1	0	0	1	4

$$\downarrow$$

x1 代替 s1

	z	a_1	x_1	x_2	s_1	s_2	s_3	
	1	1	0	0	0	0	0	0
x_1	0	1	2	1	-1	0	0	2
s_2	0	$-\frac{1}{2}$	0	$\frac{5}{2}$	$\frac{1}{2}$	1	0	1
s_3	0	0	0	1	0	0	1	4

1.2 Phase-II

	z	x_1	x_2	s_1	s_2	s_3	
	1	-3	1	0	0	0	0
x_1	0	2	1	-1	0	0	2
s_2	0	0	$\frac{5}{2}$	$\frac{1}{2}$	1	0	1
s_3	0	0	1	0	0	1	4

$$\downarrow$$

$$R0 = R0 + \frac{3}{2}R1 \tag{1}$$

	z	x_1	x_2	s_1	s_2	s_3	
	1	0	$\frac{5}{2}$	$-\frac{3}{2}$	0	0	3
x_1	0	2	1	-1	0	0	2
s_2	0	0	$\frac{5}{2}$	$\frac{1}{2}$	1	0	1
s_3	0	0	1	0	0	1	4

s1 替换 s2

	z	x_1	x_2	s_1	s_2	s_3	
	1	0	10	0	3	0	6
x_1	0	2	6	0	2	0	4
s_1	0	0	$\frac{5}{2}$	$\frac{1}{2}$	1	0	1
s_3	0	0	1	0	0	1	4

$$s_1 = 2, s_2 = 0, s_3 = 4, x_1 = 2, x_2 = 0, z = 6$$

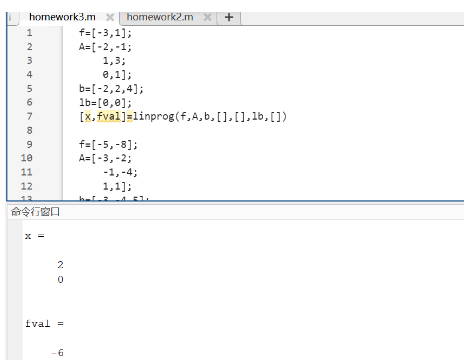

图 1: 验证结果

2 Solve following LP by Two-Phase Simplex Tableau

$$
\begin{aligned}
\text{max.} \quad & 5x_1 + 8x_2 \\
\text{s. t.} \quad & \begin{cases} 3x_1 + 2x_2 \geq 3 \\ x_1 + 4x_2 \geq 4 \\ x_1 + x_2 \leq 5 \\ x_1, x_2 \geq 0 \end{cases}
\end{aligned}
$$

2.1 Phase-I

$$
\begin{aligned}
\text{max.} \quad & -a_1 - a_2 \\
\text{s.t.} \quad & \begin{cases} 3x_1 + 2x_2 - s_1 + a_1 = 3 \\ x_1 + 4x_2 - s_2 + a_2 = 4 \\ x_1 + x_2 + s_3 = 5 \\ x_1, x_2, s_1, s_2, s_3, a_1 \geq 0 \end{cases}
\end{aligned}
$$

	z	a_1	a_2	x_1	x_2	s_1	s_2	s_3	
	1	1	1	0	0	0	0	0	0
s_1	0	1	0	3	2	-1	0	0	3
s_2	0	0	1	1	4	0	-1	0	4
s_3	0	0	0	1	1	0	0	1	5

R0-R1-R2 使得基变量系数为 0

	z	a_1	a_2	x_1	x_2	s_1	s_2	s_3	
	1	0	0	-4	-6	1	1	0	-7
s_1	0	1	0	3	2	-1	0	0	3
s_2	0	0	1	1	4	0	-1	0	4
s_3	0	0	0	1	1	0	0	1	5

x1 代替 s1

	z	a_1	a_2	x_1	x_2	s_1	s_2	s_3	
	1	$\frac{4}{3}$	0	0	$-\frac{10}{3}$	$-\frac{1}{3}$	1	0	-3
x_1	0	1	0	3	2	-1	0	0	3
s_2	0	$-\frac{1}{3}$	1	0	$\frac{10}{3}$	$\frac{1}{3}$	-1	0	3
s_3	0	$-\frac{1}{3}$	0	0	$\frac{1}{3}$	$\frac{1}{3}$	0	1	4

x2 代替 s2

	z	a_1	a_2	x_1	x_2	s_1	s_2	s_3	
	1	1	1	0	0	0	0	0	0
x_1	0	$\frac{6}{5}$	$-\frac{3}{5}$	3	0	$-\frac{6}{5}$	$\frac{3}{5}$	0	$\frac{6}{5}$
x_2	0	$-\frac{1}{3}$	1	0	$\frac{10}{3}$	$\frac{1}{3}$	-1	0	3
s_3	0	$-\frac{3}{10}$	$-\frac{1}{10}$	0	0	$\frac{3}{10}$	$\frac{1}{10}$	1	$\frac{37}{10}$

2.2 Phase-II

	z	x_1	x_2	s_1	s_2	s_3	
	1	-5	-8	0	0	0	0
x_1	0	3	0	$-\frac{6}{5}$	$\frac{3}{5}$	0	$\frac{6}{5}$
x_2	0	0	$\frac{10}{3}$	$\frac{1}{3}$	-1	0	3
s_3	0	0	0	$\frac{3}{10}$	$\frac{1}{10}$	1	$\frac{37}{10}$

$$R0 + \frac{5}{3}R1 \tag{2}$$

	z	x_1	x_2	s_1	s_2	s_3	
	1	0	-8	-2	1	0	2
x_1	0	3	0	$-\frac{6}{5}$	$\frac{3}{5}$	0	$\frac{6}{5}$
x_2	0	0	$\frac{10}{3}$	$\frac{1}{3}$	-1	0	3
s_3	0	0	0	$\frac{3}{10}$	$\frac{1}{10}$	1	$\frac{37}{10}$

$$R0 + \frac{12}{5}R2 \tag{3}$$

	z	x_1	x_2	s_1	s_2	s_3	
	1	0	0	$-\frac{6}{5}$	$-\frac{7}{5}$	0	$\frac{46}{5}$
x_1	0	3	0	$-\frac{6}{5}$	$\frac{3}{5}$	0	$\frac{6}{5}$
x_2	0	0	$\frac{10}{3}$	$\frac{1}{3}$	-1	0	3
s_3	0	0	0	$\frac{3}{10}$	$\frac{1}{10}$	1	$\frac{37}{10}$

⇓ s2 替 x1

	z	x_1	x_2	s1	s2	s3	c
	1	7	0	-4	0	0	12
s2	0	3	0	$-\frac{6}{5}$	$\frac{3}{5}$	0	$\frac{6}{5}$
x_2	0	5	$\frac{10}{3}$	$-\frac{5}{3}$	0	0	5
s3	0	$-\frac{1}{2}$	0	$\frac{1}{2}$	0	1	$\frac{7}{2}$

⇓ s1 替 s3

	z	x_1	x_2	s1	s2	s3	c
	1	3	0	0	0	8	40
s2	0	$\frac{9}{5}$	0	0	$\frac{3}{5}$	$\frac{12}{5}$	$\frac{48}{5}$
x2	0	$\frac{10}{3}$	$\frac{10}{3}$	0	0	$\frac{10}{3}$	$\frac{5}{3}$
s1	0	$-\frac{1}{2}$	0	$\frac{1}{2}$	0	1	$\frac{7}{2}$

x1=0

x2=5.

s1=7

s2=16.

s3=0

$s_1 = 0, s_2 = 37, s_3 = 0, x_1 = 0, x_2 = 5, z = 40$

```
f=[-5,-8];
A=[-3,-2;
    -1,-4;
    1,1];
b=[-3,-4,5];
lb=[0,0];
[x,fval]=linprog(f,A,b,[],[],lb,[])
```

命令行窗口

```
找到最优解。


x =

     0
     5


fval =

   -40
```

图 2: 验证结果

3 Solve following LP by Two-Phase Simplex Tableau

$$
\begin{aligned}
&\max.\ 2x_1 - x_2 + x_3 \\
&\text{s. t.} \\
&\begin{cases}
x_1 + x_2 - 3x_3 \leq 8 \\
4x_1 - x_2 + x_3 \geq 2 \\
2x_1 + 3x_2 - x_3 \geq 4 \\
x_1, x_2, x_3 \geq 0
\end{cases}
\end{aligned}
$$

问题无界解

$$\max.\ -a_1 - a_2$$

$$\begin{cases} x_1 + x_2 - 3x_3 + s_1 = 8 \\ 4x_1 - x_2 + x_3 - s_2 + a_1 = 2 \\ 2x_1 + 3x_2 - x_3 - s_3 + a_2 = 4 \\ x_1, x_2, x_3, s_1, s_2, s_3, a_1, a_2 \geq 0 \end{cases}$$

	z	a_1	a_2	x_1	x_2	x_3	s_1	s_2	s_3	c
	1	1	1	0	0	0	0	0	0	0
s_1	0	0	0	1	1	-3	1	0	0	8
s_2	0	1	0	4	-1	1	0	-1	0	2
s_3	0	0	1	2	3	-1	0	0	-1	4

$R0 = R0 - R2 - R3$ ⇓.

	z	a_1	a_2	x_1	x_2	x_3	s_1	s_2	s_3	c
	1	0	0	-6	-2	0	0	1	1	0
s_1	0	0	0	1	1	-3	1	0	0	8
s_2	0	1	0	4	-1	1	0	-1	0	2
s_4	0	0	1	2	3	-1	0	0	-1	4

⇓

	z	a_1	a_2	x_1	x_2	x_3	s_1	s_2	s_3	c
	1	$\frac{3}{2}$	0	0	$-\frac{7}{2}$	$\frac{3}{2}$	0	$-\frac{1}{2}$	1	3.
s_1	0	$-\frac{1}{4}$	0	0	$\frac{5}{4}$	$-\frac{13}{4}$	1	$\frac{1}{4}$	0	$\frac{15}{2}$
x_1	0	1	0	4	-1	1	0	-1	0	2
s_3.	0	$-\frac{1}{2}$	1	0	$\frac{7}{2}$	$-\frac{3}{2}$	0	$\frac{1}{2}$	-1	3.

7.

	z	a_1	a_2	x_1	x_2	x_3	s_1	s_2	s_3	C
	1	1	1	0	0	0	0	0	0	6
s_1	0	$-\frac{1}{14}$	$-\frac{5}{14}$	0	0	$-\frac{19}{7}$	1	$\frac{1}{14}$	$\frac{5}{14}$	5
x_1	0	$\frac{6}{7}$	$\frac{2}{7}$	4	0	$\frac{4}{7}$	0	$-\frac{6}{7}$	$-\frac{2}{7}$	$\frac{20}{7}$
x_2	0	$-\frac{1}{2}$	1	0	$\frac{7}{2}$	$-\frac{3}{2}$	0	$\frac{1}{2}$	-1	3

$\Downarrow$

	z	x_1	x_2	x_3	s_1	s_2	s_3	C
	1	-2	1	-1	0	0	0	6
s_1	0	0	0	$-\frac{19}{7}$	1	$\frac{1}{14}$	$\frac{5}{14}$	5
x_1	0	4	0	$\frac{4}{7}$	0	$-\frac{6}{7}$	$-\frac{2}{7}$	$\frac{20}{7}$
x_2	0	0	$\frac{7}{2}$	$-\frac{3}{2}$	0	$\frac{1}{2}$	-1	3

$\Downarrow$ $R0 + \frac{1}{2}R1$ $\cancel{\frac{2}{7}R2}$ $-\frac{10}{21}R2$

	z	x_1	x_2	x_3	s_1	s_2	s_3	C
	1	0	1	$-\frac{5}{7}$	0	$-\frac{3}{7}$	$-\frac{1}{7}$	$\frac{52}{7}$
s_1	0	0	0	$-\frac{19}{7}$	1	$\frac{1}{14}$	$\frac{5}{14}$	5
x_1	0	4	0	$\frac{4}{7}$	0	$-\frac{6}{7}$	$-\frac{2}{7}$	$\frac{20}{7}$
x_2	0	0	$\frac{7}{2}$	$-\frac{3}{2}$	0	$\frac{1}{2}$	-1	3

$\Downarrow$ $R0 + \frac{5}{4}R2$

	z	x_1	x_2	x_3	s_1	s_2	s_3	C
	1	5	1	0	0	$-\frac{3}{2}$	$-\frac{1}{2}$	11 / 150
s_1	0	19	0	0	1	$\frac{29}{7}$	-1	$\frac{23}{7}$
x_3	0	4	0	$\frac{4}{7}$	0	$-\frac{6}{7}$	$-\frac{2}{7}$	
x_2	0	$\frac{21}{2}$	$\frac{7}{2}$	0	0	$-\frac{7}{4}$	$-\frac{7}{4}$	$\frac{21}{2}$